Supplementary Material

Supplementary material for "Within-fruit variability of reference soluble solids sets a ceiling on near-infrared calibration performance in apple". Corresponding author: Hua Huang, huanghua@xjau.edu.cn.

This document contains Supplementary Text S1 to S5, Supplementary Tables S1 to S2 and Supplementary Figures S1 to S2, all referenced from the main text. Sections S1 and S2 give the conditions and the reproducibility record behind the Materials and methods. Section S3 lists in full the ten pre-specified operating points that §3.3 of the main text reports. Section S4 reports an analysis referenced from §3.4; no claim the paper makes depends on it, and it is given in full because the trap it illustrates is easy to miss and we walked into it ourselves. Section S5 collects the extended detail behind individual results and limitations. Table S1 belongs to Section S3; Figures S1 and S2 belong to Section S4, where they set the withdrawn day-level comparison beside the confirmed individual-level leakage in the apple data; Table S2 belongs to Section S5.7 and gives the blinded test of the inverted degradation prediction. No new experiment is reported here. Every number below is produced by the released analysis scripts and written to the corresponding archived workbook, on the same terms as the main text. Equation, section and figure numbers prefixed by S refer to this document; unprefixed ones refer to the main text.

Contents

S1 Conditions behind the attainable-ceiling bound **2**
S1.1 Zero covariance is not enough . . . 2
S1.2 The condition required, and why it is per-m . . . 2
S1.3 Exchangeability alone is not enough either . . . 2
S1.4 Status of the two conditions . . . 2
S1.5 A precise definition of terms . . . 2
S1.6 The design rule is the Spearman–Brown formula, term for term . . . 3

S2 Replication, uncertainty and cross-architecture reproducibility **3**
S2.1 Pilot and Formal . . . 3
S2.2 Repeat counts, by analysis . . . 3
S2.3 Interval construction . . . 3
S2.4 The R^2 denominator convention . . . 3
S2.5 Ratio of means, not mean of ratios . . . 4
S2.6 Cross-architecture replay . . . 4

S3 The ten pre-specified operating points in full **4**

S4 A withdrawn day-level leakage reading, and the trap behind it **5**
S4.1 The reading, and its withdrawal . . . 5
S4.2 The day-index encoding is a sparse calendar-day offset . . . 5
S4.3 Direct measurement of the confound . . . 5
S4.4 Rerun controls with targets not derived from the grouping variable . . . 5
S4.5 Claims made and not made . . . 5
S4.6 The confirmed leakage, for contrast . . . 6

S5 Extended detail behind individual results and limitations **6**
S5.1 The calyx–stem control in full . . . 6
S5.2 Why the spectral side of the apple data performs poorly . . . 7
S5.3 What the acquisition record does and does not contain . . . 7
S5.4 The two preprocessing masks . . . 8
S5.5 The semi-synthetic pre-registration and its correction . . . 8
S5.6 The retraining term, measured against a fixed predictor . . . 9
S5.7 The blinded stress test in full . . . 9

S1. Conditions behind the attainable-ceiling bound

The main text states the attainable-ceiling bound as

$$R^2 \le \frac{\sigma_a^2}{\sigma_a^2 + \sigma_f^2/m}, \qquad \mathrm{RMSE} \ge \frac{\sigma_f}{\sqrt{m}}, \tag{S1}$$

for any model that can perceive only whole-fruit information, where σ_a^2 is the between-fruit variance component and σ_f^2 the within-fruit (between-face) component of the one-way random-effects model $Y_{ij} = \mu + a_i + e_{ij}$. Two conditions are required, and §2.3 of the main text separates them: exchangeability of the five faces, and the requirement that the face deviations carry no information the predictor can see. This section gives the three constructions that show why the second condition has to be stated as mean independence rather than as zero covariance, and why it has to be required separately for each m.

S1.1. Zero covariance is not enough

Stating the second condition as $\mathrm{Cov}(a_i, \bar{e}_{i\cdot}) = 0$ constrains only linear dependence, whereas the cross-term that must vanish when a predictor g is subtracted is $\mathbb{E}[(a_i - g)\,\bar{e}_{i\cdot}]$, and for nonlinear g that expectation is not controlled by a covariance. A construction makes the gap concrete. Let a scalar whole-fruit signal be $S \sim U[0, 2\pi]$, the whole-fruit effect being its centred version $a_i = S - \pi$, and let the five face deviations be a random permutation of $\cos S, \cos 2S, \ldots, \cos 5S$. They are mutually uncorrelated and each uncorrelated with S, yet their mean is a deterministic function of S and is therefore predictable from whole-fruit information alone. Under zero covariance alone Eq. (S1) still holds, but only against predictors that are affine in the whole-fruit information, not against "any" predictor. The construction is a hypothetical chosen to violate the condition, not a model for the apple data: it makes $\bar{e}_{i\cdot}$ a function of the whole-fruit signal, so the moment estimators of the main text, whose unbiasedness requires the face deviations to be independent of a_i, are not claimed under it.

S1.2. The condition required, and why it is per-m

The condition stated in the main text is the stronger one: the mean deviation $\bar{e}_i^{(m)}$ over the m faces that serve as the reference value is mean-independent of every whole-fruit quantity a predictor can condition on,

$$\mathbb{E}\big[\bar{e}_i^{(m)} \mid a_i,\, S_i\big] = 0, \tag{S2}$$

where S_i denotes the spectrum. Holding at $m = 5$ does not imply it for a smaller subset. With e_{ij} i.i.d. centred Gaussian and $S_i = e_{i1} - e_{i2}$, for instance, $\mathbb{E}[\bar{e}_{i\cdot} \mid S_i] = 0$ while $\mathbb{E}[e_{i1} \mid S_i] = S_i/2$, so the five-face mean satisfies the condition and a single face does not. The Gaussian qualifier is needed rather than decorative: for a general i.i.d. law $\mathbb{E}[e_{i1} \mid S_i]$ equals $S_i/2$ plus an even function of S_i that need not vanish, so both identities are exact only when that even part is zero.

S1.3. Exchangeability alone is not enough either

With $a_i = e_{i1} + e_{i2}$ the face deviations are mutually uncorrelated yet perfectly correlated with their own mean, and Eq. (S1) fails.

S1.4. Status of the two conditions

Both are assumptions, not consequences of writing the model down. The first is what §3.2 of the main text tests, where a permutation test of compound symmetry cannot reject it ($p = 0.261$). The second is not testable on this dataset at all, neither the covariance version nor the mean-independence version, because it would need technical replicates on the same face; it is imposed. Under merely approximate mean independence Eq. (S1) is an approximation whose error is of the order of the residual predictable part of $\bar{e}_i^{(m)}$.

S1.5. A precise definition of terms

The m "replicates" of this paper are destructive samplings at m different positions on the same fruit, not m replicate analyses of the same sample and the same measurand. Correspondingly σ_f^2 is the spatial-subsample variability defined by the five-point scheme, pooling genuine between-face chemical heterogeneity, sampling and preparation variability, and refractometer reading error, which are inseparable here. The accurate reading of the design rule is therefore "how many approximately independent spatial subsamples per fruit are needed to estimate whole-fruit mean quality to a given precision", not "how many parallel determinations the reference analytical method must perform". The two are mathematically of the same form but correspond to different laboratory operations: the former requires moving the sampling location, the latter only re-measuring the same homogenate.

One further disclosure: the five faces per fruit were the pre-existing scheme of this collection, not a number inverted from the design rule, so the $1/m$ test of §3.2 is an in-sample check rather than a prospective application of the rule.

S1.6. The design rule is the Spearman–Brown formula, term for term

The main text inverts Eq. (S1) into $m \geq (\sigma_f^2/\sigma_a^2) \cdot R^2_{\max}/(1 - R^2_{\max})$. Write ρ for the reliability of a single determination, which is the intraclass correlation reported in §3.1; then $(1-\rho)/\rho = \sigma_f^2/\sigma_a^2$. Solving the Spearman–Brown prophecy formula $\rho_m = m\rho/[1 + (m-1)\rho]$ for a target reliability ρ^* and reading ρ^* as $R^2_{\max}$ returns the design rule term for term. Naming it is what lets the reader place what the main text adds, which is empirical and structural rather than algebraic.

S2. Replication, uncertainty and cross-architecture reproducibility

S2.1. Pilot and Formal

Everything that depends on a random split is run in two stages. Feasibility is first verified on a single development seed (Pilot). The whole analysis is then rerun on five seeds fixed in advance, {20060515, 20041210, 19810915, 2023, 2024} (Formal). Every split-dependent quantity reported in the main text is the Formal result; the Pilot values do not enter the text. A second class of quantities does not depend on any random split: the variance components and ICC, the attainable ceiling, the label-only baseline (all closed-form moment estimators), and the spectral-quality and band-extreme diagnostics (no split at all). These are seed-independent, so the original computation was retained and not rerun.

S2.2. Repeat counts, by analysis

Under each seed, 20 repeated fruit-grouped holdouts are performed ($n_{\text{primary}} = 20$), and the component count is selected by 4-fold grouped cross-validation within the training set ($K_{\text{inner}} = 4$). Each R^2 in the main apple table and in the leakage controls therefore aggregates $5 \times 20 = 100$ repeated splits. Repeated splits share data and are not independent samples. Replication counts for secondary analyses differ: the band-subset analysis uses 15 repeats per seed (75 in total), and the strong kiwifruit model, the time-series day-index controls and the semi-synthetic degradation experiment use 6 repeats per seed (30 in total each). The 20-repeat figure therefore describes the main apple table and the leakage controls, not every analysis in the paper.

S2.3. Interval construction

For every quantity that depends on a random split, confidence intervals use a two-level cluster bootstrap with seeds as clusters: seeds are resampled first, then repeats within a seed, $B = 2000$. The 20 repeats under one seed share a single random stream and are not independent samples, and an i.i.d. bootstrap systematically underestimates interval width. The reported SDs reflect only the variance introduced by splitting and modelling, not hardware jitter.

S2.4. The R^2 denominator convention

The total sum of squares in R^2 is referenced to the training-set mean, not to the test set's own mean, because the training mean is the only baseline actually available at prediction time and the test set's own mean is an oracle baseline. The direction of this choice follows from an identity,

$$\sum_i (y_i - \bar{y}_{\text{train}})^2 = \sum_i (y_i - \bar{y}_{\text{test}})^2 + n(\bar{y}_{\text{test}} - \bar{y}_{\text{train}})^2, \tag{S3}$$

by which the training-mean denominator is never smaller, so this convention yields an R^2 that is never lower than the test-mean convention. Replaying every split used in the paper without refitting, the ratio ρ of the two denominators has mean 0.999 for the kiwifruit runs and 0.994–0.996 for the apple main table. The difference between the conventions is $(1 - R^2)(1/\rho - 1)$ and so scales with $1 - R^2$: on the kiwifruit data it is 0.0002 in R^2; on the apple data, where every R^2 is below 0.12, it is 0.0036 (face-level target) and 0.0058 (fruit-mean target) on average over the 100 splits, reaching 0.033 and 0.055 respectively in the most extreme single split. Converting each split of the paired attenuation test of §3.2 to the test-mean convention moves the ratio from 0.564 to 0.561 and its seed-clustered interval to [0.526, 0.595], which still contains the prediction 0.5816 and still excludes the ICC 0.477. Every comparison the main text draws between a measured score and the attainable ceiling, and equally the six point predictions and the design rule, is therefore approximate to that order rather than an exact identity; the per-split conversion is in the companion workbook.

S2.5. Ratio of means, not mean of ratios

Every ratio-valued quantity (attenuation ratio, split-inflation factor, RMSE ratio) is computed as a ratio of means $\mathbb{E}[X]/\mathbb{E}[Y]$ rather than a mean of ratios $\mathbb{E}[X/Y]$. With a denominator as small as $R^2 \approx 0.06$, the latter is pulled up appreciably by a few very small denominators. We do not claim a general direction for this discrepancy; for dependent X and Y neither ordering holds universally. It is reported as measured: on the present data the mean of ratios reaches 2.28 against 1.41 for the ratio of means (see the companion workbook). The decisive reason for the choice is nonetheless structural rather than empirical, in that the theoretical predictions being tested are themselves ratios of expectations, so only the ratio of means is of the same form.

S2.6. Cross-architecture replay

All models are PLS and closed-form moment estimators with no tensor operations, and computation runs on CPU. Given the seed, a rerun on the same architecture and BLAS implementation returns the reported values bit for bit; across architectures it does not. The component count is chosen from an inner grouped cross-validation MSE curve that is nearly flat in the weak-signal configurations (relative differences of order 10^{-3}), so floating-point differences between BLAS implementations can move the selection, and a moved selection shifts the aggregate. Replaying the analysis on Apple silicon under the original seeds left 181 of 237 checked quantities unchanged. Of the 56 that moved, 32 are in the apple spectral analyses (R^2 between 0.015 and 0.17, where the inner curve is flattest), 18 in the time-series controls of §S4 (including the interval-width ratio of Fig. S1, 479 against 496), and six in the semi-synthetic experiment, where no shift exceeds 1.5×10^{-4} in R^2 or 0.014 percentage points in the relative deviation. The largest shift among the quantities recorded at full precision is in the upper confidence limit of the split-inflation factor for raw spectra against face-level targets: 1.55 against the reported 1.71, or −9.3%; the lower limit of the 900–1001 nm band-subset interval moves by a larger fraction (0.007 against the recorded 0.008) but only in its third decimal. No statistical conclusion reverses under the replay: the attenuation-ratio interval still contains the predicted 0.5816 and still excludes the ICC of 0.477, and all four leakage intervals still have lower limits above 1. The dependence is removable rather than merely disclosed. The component count selected for every configuration is released with the model records, and holding it fixed at the recorded value reproduced one repetition of the full-spectrum band-subset run (seed 2024) to the sixth decimal place on the other architecture; that check was made once, not for every configuration. The software environment of the machine that produced the reported numbers was not recorded at the time; the pinned environment released with the code is the one in which this replication was run, and is labelled as such. The intermediate products of the replay were not retained, so it can be repeated from the scripts and seeds but not rechecked quantity by quantity.

S3. The ten pre-specified operating points in full

Section 3.3 of the main text states that ten configurations were fixed in advance and that all ten are reported. They are listed here in full, ordered by R^2. Nothing was added or removed after results were seen; the list is the `CONFIGS` constant of the operating-point scan script `A12opscan` (v18). Each row is the mean over the five fixed seeds. "Pooled component" is the part of RMSEP corresponding to the dispersion of predictions for the same fruit across devices; "share" is that component's share of the mean squared prediction error.

Table S1
All ten operating points of the scan behind Fig. 3(b) of the main text.

ID	Preprocessing	Range (nm)	Comp.	R^2	RMSEP	Pooled comp.	Share (%)
C1	raw	≥402	2	0.097	2.207	0.457	4.3
C3	SNV	≥402	4	0.251	2.010	0.562	7.8
C2	raw	≥402	6	0.267	1.988	0.495	6.2
W	SNV	≥402	10	0.394	1.808	0.497	7.6
C5	1st deriv.	≥402	6	0.451	1.720	0.514	8.9
C6	1st deriv.	≥700	6	0.580	1.506	0.524	12.1
C4	SNV	≥700	10	0.718	1.232	0.487	15.6
C8	SNV + 1st deriv.	≥800	CV	0.792	1.060	0.516	23.7
C7	2nd deriv.	≥700	10	0.800	1.039	0.508	23.9
S	1st deriv.	≥700	CV	0.820	0.986	0.490	24.7

W and S are the two operating points of the earlier two-point analysis, re-run here under the scan's own fold rule; their values under that rule ($R^2 = 0.394$ and 0.820) are close to but not identical with the values quoted in the main text from the deterministic grouped k-fold analysis (0.393 and 0.819), and the two sets must not be interchanged.

S4. A withdrawn day-level leakage reading, and the trap behind it

S4.1. The reading, and its withdrawal

On time-series hyperspectral data (post-harvest ripening sequences of five species, 3,405 VIS spectra) we initially observed what appeared to be a deeper layer of leakage. Predicting the imaging day index of a single SNV-preprocessed spectrum gives $R^2 = 0.887$ (RMSE 4.31) when grouped by fruit, and falls to 0.705 (RMSE 6.85) when grouped by acquisition day. By species the drop is more dramatic still: kiwifruit falls from 0.980 (RMSE 1.61) to −0.14 (RMSE 10.81), and avocado from 0.964 (RMSE 2.86) to 0.844 (RMSE 6.16). The reading at the time was that this is day-level leakage caused by individuals imaged on the same day sharing illumination and ambient conditions.

That reading does not survive a control test and we withdraw it here. The first suspicion comes from the intervals themselves: the fruit-grouped R^2 barely moves across the five seeds (kiwifruit [0.979, 0.982]), whereas the same quantity under day grouping spans [−0.81, +0.37] (Fig. S1). A genuine leakage magnitude should not behave that way. The problem is that the target of that analysis is itself derived from the grouping variable: the imaging day index is numbered from the acquisition-day label (27 acquisition days in one-to-one correspondence with 27 index values), so holding out a whole day also removes an entire level of the target, and the target values of the test samples never appear in the training set. The performance drop therefore mixes two things: removing shared same-day conditions (leakage) and extrapolating to target values outside the training range (a mechanical degradation unrelated to leakage).

S4.2. The day-index encoding is a sparse calendar-day offset

The correspondence just stated is one-to-one but not dense, and the encoding is stated here explicitly because the pairing of "27 acquisition days" with a maximum index of 37 is easy to read as an inconsistency. On the kiwifruit subset the day index is a calendar-day offset from the first acquisition: it takes 27 distinct values with minimum 0 and maximum 37, namely $\{0, 1, \ldots, 9\} \cup \{21, 22, \ldots, 37\}$, so the eleven calendar days 10–20 carry no imaging and are absent from the index. The map between acquisition day and index is a bijection on those 27 values. The count of levels, not the numerical range, is what the argument above uses.

S4.3. Direct measurement of the confound

We split the 30 holdouts by whether the held-out set contains a terminal date (the minimum day index of 0 or the maximum of 37). Those without a terminal date ($n = 12$) have mean R^2 +0.715, range [+0.147, +0.880]; those with one ($n = 18$) have mean R^2 −0.709, range [−4.392, +0.892]. The difference is +1.424, with a two-level seed-clustered bootstrap 95% confidence interval of [+0.592, +2.574] that excludes 0. The size of the drop is thus governed mainly by how far one extrapolates, not by how much shared same-day condition was removed.

S4.4. Rerun controls with targets not derived from the grouping variable

To separate the two, we reran the same comparison using destructive measurements in that dataset that are not derived from the grouping variable: fruit firmness ($n = 694$ labelled spectra) and storage duration ($n = 717$). Both take several values within a single acquisition day, so day grouping no longer removes an entire level of the target. The criterion was fixed before the run: day-level leakage is accepted only if the ratio of day-grouped to fruit-grouped RMSE is ≥ 1.5. The six combinations are the two targets crossed with three evaluation levels (kiwifruit, avocado and the pooled set), not three species: of the five species in this dataset, only kiwifruit and avocado carry enough labelled records for a per-species five-seed evaluation (236 and 238 for firmness, 251 and 246 for storage duration), whereas kaki, mango and papaya carry 60–80 each and enter only through the pooled level. Only 2 of those 6 combinations meet the criterion, and both occur on storage duration, which, while not a function of acquisition day, remains strongly correlated with it; tightening the criterion to require the lower bootstrap 95% confidence limit also to be ≥ 1.5 leaves only 1. On fruit firmness, the target most independent of acquisition day, all three variants fail (kiwifruit 0.74, avocado 1.23, pooled 1.08), and kiwifruit even points the other way.

S4.5. Claims made and not made

We therefore do not claim that the time-series data exhibit a day-level leakage mechanism independent of individual-level leakage. The episode is reported because it constitutes a reproducible validation-protocol trap in its own right: when the prediction target is derived from the grouping variable, grouping by that variable manufactures a leakage-like performance drop out of nothing. Such targets are not rare in practice (imaging batch, scan order or storage-stage index used as a proxy label), and the size of the drop alone cannot distinguish them from genuine leakage: one has to do what this section does, and rerun with a reference determination independent of the grouping variable.

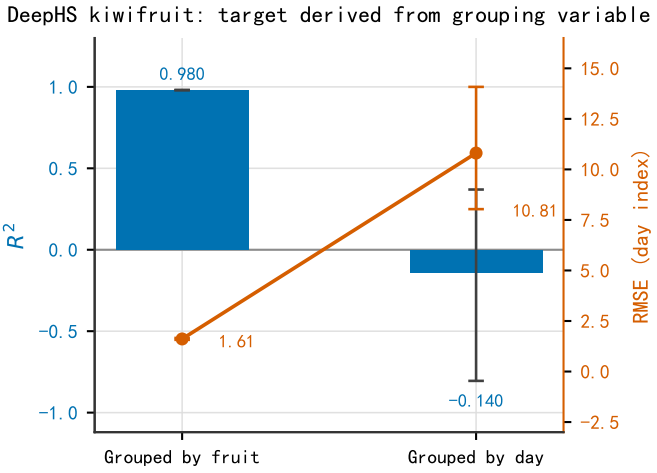


Figure S1: The withdrawn day-level comparison (time-series hyperspectral data, kiwifruit): the drop produced by grouping by acquisition day, attributed after the control test to a target derived from the grouping variable rather than to day-level leakage. Left axis R^2 (blue bars), right axis RMSE (red line); error bars are two-level seed-clustered bootstrap 95% confidence intervals ($n_{\text{seeds}} = 5$, 6 repeated holdouts per seed). The interval under day grouping is 1.1749 wide, 479 times the width under fruit grouping (0.002455; the extra digits let the ratio be recomputed from the two numbers printed here; a replay on another architecture returns 0.002369 and a ratio of 496, §S2), and that instability is itself evidence for the attribution above. All values are Formal five-seed results.

S4.6. The confirmed leakage, for contrast

The leakage that does survive its controls is the individual-level one in the apple data, reported in §3.4 of the main text and drawn in Fig. S2. It is placed here so that the confirmed and the withdrawn comparison can be read side by side.

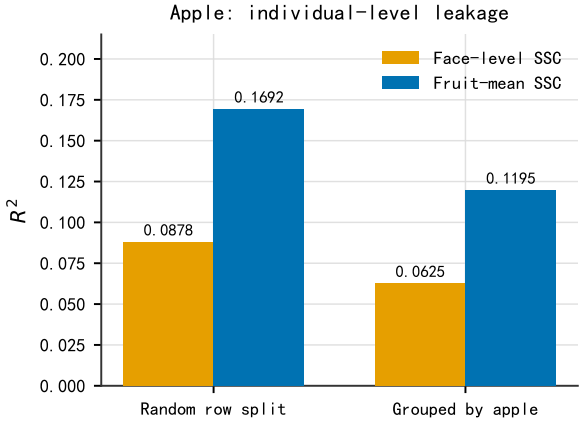


Figure S2: Genuine individual-level leakage in the apple data: a row-wise random split inflates performance relative to grouping by fruit. The bars plot the raw-preprocessing comparison for the face-level and the fruit-mean SSC target (factors 1.41 and 1.42); the first-derivative comparison gives 1.45 and 1.46 and is reported in the text of §3.4 of the main text rather than drawn here, so the range quoted there, 1.41–1.46×, spans all four combinations. All values are Formal five-seed results.

S5. Extended detail behind individual results and limitations

S5.1. The calyx–stem control in full

An immediate objection to the within-fruit variability reported in §3.1 of the main text is that the bottom (calyx end) of a fruit is known to be sweeter than the top (stem end), so the variability might be nothing but a restatement of that gradient. The gradient is confirmed: the bottom exceeds the top by 0.405 °Brix (paired $t = 4.42$). Its magnitude is nonetheless only 36% of the median within-fruit SD, and the fixed effect of face position explains just 0.95% of the within-fruit variability. A cleaner control uses the three side faces only: they

lie on the same equatorial band, are anatomically equivalent, and contain no stem–calyx gradient at all. The between-face variance estimated from them is 2.4396, i.e. 95.6% of the value 2.5515 estimated from all five faces, and the median range across just those three faces is still 1.80 °Brix. To be precise: the face-position effect is not statistically zero (permutation test $p < 0.001$), but its explanatory power is very weak (0.95%). The variability therefore has almost no structure in the spatial variables this dataset records. That is not to say it has no biological structure, since it may relate to unrecorded factors such as sun exposure or canopy position, and precisely for that reason standardising the sampling position alone will not eliminate it. Substantial spatial dynamics of sugars and cell state within apple flesh have direct cytological evidence in watercored fruit; watercore is a physiological disorder, so that evidence is not extended to sound fruit. It is compatible with the dispersion observed at the reference-value level, but because this dataset has no technical replicates on the same face it cannot be used to attribute that dispersion to biological heterogeneity.

S5.2. Why the spectral side of the apple data performs poorly

Under fruit-grouped holdout, the R^2 for whole-fruit mean SSC is only about 0.11. That figure comes from two independently written evaluation implementations. The main implementation gives 0.118 (95% CI [0.109, 0.127], component count searched over 1–20, 20 repeats per seed). The other, with a different component grid and repeat count (2–24 in steps of 2, 15 repeats per seed), gives 0.109 (95% CI [0.095, 0.121]). The two intervals overlap strongly, which supports only "the conclusion is insensitive to implementation details and to the random order of splitting"; it does not constitute replication on independent data. Either way, both are far below that dataset's five-face-mean ceiling of 0.820 (the one that matches this target) and the conclusion is unaffected. Three diagnostics failed to support the candidate explanations.

1. Not the splitting scheme. A row-wise random split gives only 0.169.
2. Not clipping or band selection. The archived spectra were min–max scaled to $[0, 1]$ spectrum by spectrum: every one of the 3,228 archived spectra has a minimum of exactly 0 and a maximum of exactly 1, so detector saturation cannot be read from their values. The signature of clipping, a plateau of ten or more bands tied at the maximum, occurs in 33 spectra (1.0%); in the other 3,195 the maximum is reached at a single band, whose position has 5th, 50th and 95th percentiles of 708.3, 805.5 and 814.6 nm. A value above 0.99 or below 0.01 therefore marks a band lying within 1% of the spectrum's own maximum or minimum, not a saturated band. Under the second implementation the full spectrum gives 0.109; excluding the bands at which more than 1% of spectra lie within 0.01 of either end of the scaled range is worse, at 0.077; and 900–1001 nm alone is lower still, at 0.015.
3. No sign of a general modelling failure. The same code and protocol give $R^2 = 0.827$ for DM and $R^2 = 0.861$ for SSC on the kiwifruit data. That protocol is structurally identical to the apple analysis, fruit-grouped repeated holdout with 4-fold inner grouped cross-validation for component selection, but uses 6 repeats per seed against 20 for the apple main table and 15 for the band subsets. It is also a different evaluation regime from the $R^2 = 0.819$ obtained under 5-fold grouped cross-validation in §3.3 of the main text; the two must not be mixed.

The single band most correlated with SSC is at 678 nm ($|r| = 0.154$), in the chlorophyll absorption and red-edge region, which is a colour–maturity proxy rather than direct sugar absorption. The reasonable attribution is twofold. Reflectance geometry on intact fruit yields a surface-weighted sampling whose effective depth is typically shallow and varies with wavelength, geometry and fruit structure, whereas the refractometer measures juice from destructively sampled flesh. In addition, 391–1001 nm covers only the very weak third overtone, the strong sugar absorption bands lying outside the range.

S5.3. What the acquisition record does and does not contain

The source of the within-fruit variability cannot be fully decomposed, because this dataset has no technical replicates on the same sampling face. So long as the face-level departures are approximately independent across faces, genuine spatial chemical heterogeneity and sampling-and-preparation error both decay as $1/m$ alike: the ceiling and the design rule are unchanged word for word, and only the accompanying recommendation changes, in that besides raising the number of replicates one should also standardise the sampling procedure. That independence is precisely what the compound-symmetry permutation test of §3.2 examines ($p = 0.261$). Conversely, a systematic procedural bias acting at the whole-fruit level is by definition absorbed into σ_a^2, which can only raise the ICC and make the ceiling look more permissive; it cannot make the conclusions optimistic. Whether σ_f^2 is biological heterogeneity or procedural error therefore affects where the improvement effort should go, not whether the design formula holds.

To let the reader judge the boundary of this reading, what the acquisition record does and does not contain is stated here. It does define the sampling faces (top, bottom, and three sides taken every 120° counterclockwise

around the equatorial band) and the procedure that the refractometer is read after destructive sampling. It does not record the mass of tissue taken per face, whether each face was juiced separately, or whether the readings were temperature-compensated. Those three are precisely what would be needed to separate genuine spatial chemical heterogeneity from sampling-and-preparation error, so no claim is made about their relative shares.

S5.4. The two preprocessing masks

Two feature masks in the pipeline are built from the whole dataset and only then fed to the grouped holdouts. The first is the band mask behind the band-subset diagnostic (formal-pipeline script `A6formal`, v18), which retains wavelengths whose combined under/over-range rate is $\leq 1\%$, that is, wavelengths at which no more than 1% of spectra lie within 0.01 of either end of the scaled range. The second is the missing-channel mask for the kiwifruit model (the same script), which drops wavelength columns absent for some devices. Both are target-blind: they read only the spectra, never the reference values, and dropping channels fold-by-fold would give the folds different feature spaces and make them incomparable. Under a strict no-clean-separation criterion, however, they are still decisions taken with held-out spectra in view. A published route removes the trade-off rather than bounding it: compute the statistics once on the whole set, then subtract each validation partition's contribution fold by fold. It was not taken here, and is recorded as the remedy for a future run. The affected quantities are reported rather than re-derived: the band-subset diagnostic, and the kiwifruit R^2 of 0.827/0.861 used as the clean baseline in the semi-synthetic experiment.

For the band mask the effect was measured rather than bounded. That mask is not a no-op: 76 of the 346 wavelengths have an under/over-range rate within 0.5 percentage points of the 1% threshold, and recomputing the mask inside each of the 75 outer training partitions of the band-subset run retains between 68 and 146 wavelengths (median 99), against 100 for the whole-data mask. Re-running the band-subset diagnostic with the partition-local mask gives $R^2 = 0.081$ (seed-clustered 95% interval [0.074, 0.087]) against 0.075 with the whole-data mask, both on the replay architecture of §S2, where the reported 0.077 replays as 0.075 and the full spectrum, run in the same pass, gives 0.106. The ordering full spectrum > band subset, and with it the conclusion that band restriction does not repair the apple performance, is unchanged; the whole-data mask biases the subset downwards, not upwards.

For the kiwifruit mask the question can be settled rather than merely bounded. After the rows lacking a reference value are dropped, the spectral block contains no missing value at all: the mask removes 0 of 222 wavelength columns. A mask recomputed inside a training split can only ever keep a superset of the whole-data mask, since fewer rows mean fewer opportunities for a column to be missing. Once the whole-data mask keeps every column, any fold-local mask is therefore necessarily identical, for every conceivable split. This was also checked directly on the splits actually scored: recomputing the mask inside each of the $5 \times 6 = 30$ outer training splits of the reported kiwifruit run reproduces it 30/30 times. Its position relative to the split therefore cannot affect the evaluated feature space, the predictions or the scores. Separately, one exploratory script selects a preprocessing method by held-out R^2; no number from that selection enters the paper, and it is flagged here because the released code contains it.

S5.5. The semi-synthetic pre-registration and its correction

The criteria of the semi-synthetic experiment were all written down and archived before it was run. In the interest of full disclosure, the version used is the second version of that pre-registration. The first (Appendix G of the protocol) set the ceiling against a variance the model could in fact perceive, and its own hard-falsification criterion failed; it is kept verbatim rather than deleted. The present version corrects that setting, with no threshold changed, and its four design targets were fixed before the corresponding replicate levels had been measured.

The pre-registered agreement criterion, stated on the absolute value, is $|R^2_{\mathrm{obs}}/R^2_{\mathrm{pred}} - 1| \leq 15\%$. Passing it already keeps the measurement below the structural ceiling at the clean baselines reported here: R^2_{pred} sits below the ceiling by the factor R^2_{clean}, and that margin is wider than the tolerance. This is an arithmetic consequence of those baselines rather than an identity; for a clean baseline close to one the same criterion would not guarantee it. The observation that the measurement stays below the ceiling at all 13 replicate levels is therefore consistent with the declared injected variance but does not by itself establish it: staying below the ceiling is compatible with more than one injected magnitude. What the injection realises rests on the implementation and the pre-registered parameters, not on this observation, which is in any case not a test independent of the agreement criterion.

One further property of the construction is recorded here because it explains the regularity of Fig. 4(b) of the main text. Within each seed the twelve finite levels of m share the same standardised noise realisation, scaled by $1/\sqrt{m}$, and the same holdout partitions, because the random stream is seeded once per (target, m, seed) unit and consumes the same draws at every finite m. The levels are therefore a paired design: each point is an independent retraining over 30 partitions, but partition noise cancels between levels, so the monotone shrinkage of the deviation is a property of the pairing, not of any smoothing. The clean baseline at $m = \infty$ draws no noise and so runs on

its own partitions; the asymptote of the deviation, though not its shape, therefore carries a partition component of the order of the baseline's own interval half-width (0.5% for DM, 0.2% for SSC).

S5.6. The retraining term, measured against a fixed predictor

Equation (5) of the main text carries R^2_{clean} through the injection unchanged. That is exact for a predictor fixed before the injection, one whose error is then uncorrelated with the injected noise, whereas the procedure evaluated here refits on the noisy labels and selects its component count with them. The gap between the two was measured rather than argued. Each repeat was rerun a second time with the components and the coefficients chosen on the clean labels of the same training partition, everything else held fixed: same seed, same noise realisation, same split, same R^2 denominator. The difference $\Delta = R^2_{\text{fixed}} - R^2_{\text{retrained}}$ is the retraining term.

Δ is positive at all twelve finite levels for both targets, with cluster bootstrap intervals (2,000 resamples over units, as elsewhere) excluding zero throughout. It is largest where the injected noise is largest and shrinks monotonically: for DM from 0.0039 at $m = 1$ (95% CI $[0.0032, 0.0046]$) to 0.0002 at $m = 20$, and for SSC from 0.0038 ($[0.0032, 0.0043]$) to 0.0002; pooled over the twelve levels, 0.00123 $[0.00098, 0.00151]$ for DM and 0.00112 $[0.00088, 0.00138]$ for SSC. Refitting therefore does cost performance, in the direction that makes Eq. (5) optimistic for the retrained procedure and the design rule of Eq. (4) conservative.

It does not account for the whole deviation. In the same paired rerun the retrained arm sits below Eq. (5) by -1.68% (DM) and -1.58% (SSC) at $m = 1$ and by -0.66% and -0.12% at $m = 20$, which differs from the values reported in the main text by at most 0.01 percentage points, within the single-architecture tolerance of §S2; with the predictor fixed the shortfall is still -0.97% and -0.92% at $m = 1$, and -0.63% and -0.09% at $m = 20$. The retraining term therefore contributes 0.72 and 0.66 percentage points at $m = 1$ and 0.03 at $m = 20$, and the rest is present for a predictor that never saw the injected noise. The deviation vanishes at $m = \infty$ by construction, because R^2_{clean} is taken as the measured $m = \infty$ value of the retrained arm; that level calibrates the curve rather than testing it. Two limits are worth stating. The fixed arm is refitted on the same training partition with clean labels, so it is one realisation of "fixed before the injection", not the only one; and this is a measurement of what the invariance assumption costs on these data under this evaluator, not a proof of the assumption. The comparison is generated by `code/A16retrainterm.py` and the per-repeat values are released with the workbook.

Because the two arms run within the same repeat on the same machine, Δ is a paired difference and is not affected by the component-count flips that platform floating-point differences can cause (§S2); the absolute R^2 of either arm is a single-architecture value in the sense set out there.

S5.7. The blinded stress test in full

Four blinded targets were specified in advance (the m at which each lies had no measured result at the time of specification), inverted for the required replicate count, and only then measured (Table S2). What is inverted is the semi-synthetic degradation prediction, not the design rule of the main text. Two things differ. In the semi-synthetic setting the model perceives the entire clean label, so the denominator carries σ^2_y rather than σ^2_a. The clean baseline itself also falls short of 1, and this must be subtracted:

$$m_{\text{req}} \;=\; \left\lceil \frac{\sigma^2_f}{\sigma^2_y} \cdot \frac{R^2_t}{R^2_{\text{clean}} - R^2_t} \right\rceil. \tag{S4}$$

The two expressions have the same form, both inverting "explained variance over total variance", and differ only in which variance appears in the denominator and in the fact that the clean baseline of real data is not 1.

Table S2
Blinded test of the inverted semi-synthetic degradation prediction, Eq. (S4), not of the main-text design rule. The shortfall is the measured R^2 minus the target value.

Target	R^2 target	Inverted m_{req}	Measured R^2	Shortfall
DM	0.75	6	0.7536	+0.0036
DM	0.78	9	0.7750	−0.0050
SSC	0.75	4	0.7578	+0.0078
SSC	0.80	7	0.7989	−0.0011

Of the four blinded targets, two were met and two fell marginally short (all shortfalls within ± 0.008). This is the same phenomenon as the negative deviation reported in the main text: the design rule accounts only for the structural attenuation caused by reference noise and not for the additional loss incurred when the model is retrained on noisy labels. The status of the rule is therefore restated accurately as "a lower bound on the required number of replicates", together with an advisory margin rule: discount R^2_{clean} by 1% before inverting Eq. (S4). The

expression actually inverted in this operational form is Eq. (S4) carrying that empirical margin, not the main-text design rule itself.

Under that correction (the per-target inversion and lookup are in the companion workbook), two things must be distinguished. First, two of the four blinded targets have their m_{req} unchanged (DM 0.75 stays at $m = 6$, SSC 0.75 stays at $m = 4$). Both already met their targets before the correction (0.7536, 0.7578) and therefore do not test the margin rule at all. Second, only two targets are actually raised by the rule, exactly the two that previously fell short. SSC 0.80 rises from $m = 7$ to $m = 8$, where the measured value is 0.8062, turning a shortfall into a pass; this is the one occasion on which the margin rule is directly verified by measurement. DM 0.78 rises from $m = 9$ to $m = 11$, a level that lies outside the grid $(1\text{–}10, 14, 20, \infty)$ and therefore has no measured value. No interpolation is performed; only the neighbouring levels are reported: $m = 10$ measures 0.7794 (still 0.0006 short) and $m = 14$ measures 0.7911 (meets the target). The rule is thus directly verified in exactly one case and merely bracketed in the other, with the remaining two targets not participating in the test. The 1% figure is accordingly an empirical margin induced from these four points, not a universal constant; its evidential basis is one direct verification plus one bracketing, and it must be recalibrated on other data.